

🌟 RS003 Week 12 — English Worksheet

Topic: Twinkling Stars and Drifting Clouds · **Course:** Pygame Turret · **Time:** about 45 minutes

This week your background **moves**. Stars twinkle at different brightnesses and clouds drift slowly across the screen, wrapping back around when they leave. You store each star and cloud as a small **dictionary** of facts and loop over a **list** of them every frame.

💬 Words to know: **list**, **dictionary**, **drift**, **wrap-around**, **brightness**

1 · Predict 🌟

Read each piece of code. Before you run it, write what you think will happen.

```
stars = []
for i in range(80):
    stars.append({"x": random.randint(0, WIDTH), "y": random.randint(0, HEIGHT - 100), "phase": random.randint(0, 60)})
```

This builds a list of 80 stars. What three facts does each star remember?


```
cloud["x"] = cloud["x"] - cloud["speed"]
if cloud["x"] < -cloud["size"]:
    cloud["x"] = WIDTH + cloud["size"]
```

Each cloud moves left a little every frame. What happens when a cloud goes off the left edge?


```
brightness = 100 + ((frame + star["phase"]) * 4 % 156)
pygame.draw.circle(screen, (brightness, brightness, brightness), (star["x"],
star["y"]), 2)
```

The brightness changes every frame and each star has a different **phase** . What do the stars look like together?

2 · Run It

 **Type the example, run it, and watch it breathe**

```

import pygame
import random
pygame.init()

WIDTH, HEIGHT = 800, 600
screen = pygame.display.set_mode((WIDTH, HEIGHT))
clock = pygame.time.Clock()

stars = []
for i in range(80):
    stars.append({"x": random.randint(0, WIDTH), "y": random.randint(0, HEIGHT - 100), "phase": random.randint(0, 60)})

clouds = []
for i in range(5):
    clouds.append({"x": random.randint(0, WIDTH), "y": random.randint(40, 200), "speed": random.uniform(0.3, 1.0), "size": random.randint(40, 80)})

frame = 0
running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False

    screen.fill((20, 30, 80))

    for star in stars:
        brightness = 100 + ((frame + star["phase"]) * 4 % 156)
        pygame.draw.circle(screen, (brightness, brightness, brightness), (star["x"], star["y"]), 2)

    for cloud in clouds:
        cloud["x"] = cloud["x"] - cloud["speed"]
        if cloud["x"] < -cloud["size"]:
            cloud["x"] = WIDTH + cloud["size"]
            cloud["y"] = random.randint(40, 200)
        cx = int(cloud["x"])
        pygame.draw.circle(screen, (200, 200, 220), (cx, cloud["y"]), cloud["size"] // 2)
        pygame.draw.circle(screen, (200, 200, 220), (cx + 20, cloud["y"] - 10), cloud["size"] // 2)
        pygame.draw.circle(screen, (200, 200, 220), (cx - 20, cloud["y"] - 5), cloud["size"] // 2)

```

```
pygame.display.flip()  
frame = frame + 1  
clock.tick(60)
```

```
pygame.quit()
```

Watch for 10 seconds. Describe what the stars do and what the clouds do:

3 · Spot the Bug 🐛

Each block below was meant to do something but is broken. Fix it, then explain why the original was wrong.

Bug A — All the stars are supposed to twinkle at different times, but instead they all flash on and off together.

```
stars.append({"x": random.randint(0, WIDTH), "y": random.randint(0, HEIGHT - 100),  
"phase": 0})
```

Hint: every star has the same `phase`, so they share the same beat.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — Clouds should keep flowing forever, but once they leave the left side they never come back and the sky goes empty.

```
cloud["x"] = cloud["x"] - cloud["speed"]  
pygame.draw.circle(screen, (200, 200, 220), (int(cloud["x"]), cloud["y"]),  
cloud["size"] // 2)
```

Hint: there is no wrap-around to send the cloud back to the right side.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The program crashes when drawing a cloud because a position has a decimal point.

```
cloud["x"] = cloud["x"] - cloud["speed"]
pygame.draw.circle(screen, (200, 200, 220), (cloud["x"], cloud["y"]), cloud["size"]
// 2)
```

Hint: `speed` is a decimal, so `cloud["x"]` becomes a decimal. Positions must be whole numbers.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Modify It

Add more clouds at different speeds

Change `range(5)` to make more clouds. The speeds and sizes are already random, so they will all differ.

Write the new cloud count you chose and how the sky looks now:

Give stars a few different colours

Most stars are white, but make some of them slightly yellow or blue.

Describe how you decided which stars get which colour:

Hint: you can store a `"color"` fact in each star's dictionary, picked with `random.choice(...)` when you build the list.

5 · Make It 📷

Build a living background behind your turret

Build a scene with a moving background — twinkling stars, drifting clouds, or your own idea like falling snow — placed behind your turret from earlier weeks.

Send a **photo or video** of your living scene, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I stored each star as a dictionary with ...

I looped over the list every frame to ...

The clouds drift because I ...

They wrap around the screen by ...

One tricky moment was when ...

If I had more time, I would ...

6 · Record Your Walkthrough 🎥

Take a video of your living background. Talk through it like you are teaching someone. Try to use these words: **list**, **dictionary**, **drift**, **wrap-around**, **brightness**.

1. Show the stars twinkling and clouds drifting.
2. Read the line that builds one star out loud and name its facts.
3. Point at the wrap-around code and explain it.
4. Change the cloud count live and show the difference.

Write what you will say in your video. Plan it here before you record.

Submit 

Send this worksheet + your photo or video to teacher on KakaoTalk.